

Kosuke Ukita

PhD student (D1), full-time

Kyushu Institute of Technology

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EDUCATION

Apr. 2026 - Present	PhD Kyushu Institute of Technology Department of Computer Science and Systems Engineering, Graduate School of Computer Science and Systems Engineering <u>Thesis: ""</u> GPA: - / 4.0
Apr. 2024 - Mar. 2026	MSc Kyushu Institute of Technology Department of Creative Informatics, Graduate School of Computer Science and Systems Engineering <u>Thesis: "A Study on Sensor Foundation Models via Joint Training using Flow Matching"</u> GPA: 3.6 / 4.0
Apr. 2020 - Mar. 2024	BSc Kyushu Institute of Technology Department of Artificial Intelligence, Computer Science and Systems Engineering <u>Thesis: "Proposal of Representation-Conditional Latent Diffusion Models: Hematoma Classification in Brain CT Images"</u> GPA: 3.7 / 4.0
Apr. 2017 - Mar. 2020	High School Diploma Kanonji Daiichi High School (Designated as a Super Science High School) Science and Mathematics Course GPA: 4.9 / 5.0

WORK EXPERIENCE

Dec. 2023 - Mar. 2026	Research Assistant (RA) Supervisor: Tsuyoshi Okita Researched and Developed foundation models
Apr. 2024 - Jun. 2026	Teaching Assistant (TA) Lecturer: Tsuyoshi Okita - Deep Learning - Deep Learning Exercise - Deep Learning Advanced I
Oct. 2023 - Nov. 2023	OPTiM Corporation Research Intern / Part-time Research Assistant Research and developed AI technologies for agricultural DX

SOCIAL EXPERIENCE

Jun. 2026	Conference - Student Volunteer (SV) The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026) · Hong Kong, China - Registration desk - Session support - Excursion support - On-site support
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PUBLICATIONS

[1] **Efficient Inference for Flow Matching via Unified Path CFG**
 Kosuke Ukita, Tsuyoshi Okita
 The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)
 June 2026 · Hong Kong, China

Refereed

Oral

[2] **High-Performance Self-Supervised Learning by Joint Training of Flow Matching**
 Kosuke Ukita, Tsuyoshi Okita
 The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)
 May 2026 · Tangier, Morocco

Refereed SpotLight (top 3%)

[3] **Foundation Models to Tackle Activity Recognition in Unknown Domain: Sussex-Huawei Locomotion Challenge 2025 Task 2**
 Tsuyoshi Okita, Kosuke Ukita, Asahi Miyazaki, Daichi Kubota, Jukichi Ota, Naoki Kagiyaama, Asahi Nishikawa, Daichi Nagayasu, Syunya Tomitaka, Daisuke Nozaki, Yuki Odo, Raku Yamashita, Xiaolong Ye, Huayu Gao, Kazuki Okahashi, Koki Matsuishi, Masaharu Kagiyaama, Kodai Hirata, Haruki Kai, Lin Wang, Hristijan Gjoreski, Mathias Ciliberto, Paula Lago, Kazuya Murao, Daniel Roggen
 Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '25), pp. 983–991
 October 2025 · Espoo, Finland

Refereed Oral

[4] **Multimodal Foundation Model for Cross-Modal Retrieval and Activity Recognition Tasks**
 Koki Matsuishi, Kosuke Ukita, Tsuyoshi Okita
 International Journal of Activity and Behavior Computing (IJABC), Care XDX Center, Kyushu Institute of Technology, Vol. 2025, No. 3, pp. 1-25
 November 2025 · Kitakyushu, Japan

Refereed Oral

[5] **Energy-Efficient High-Performance Representation Learning for Sensor Data via Flow Matching**
 Kosuke Ukita, Tsuyoshi Okita
 The 7th International Symposium on Neuromorphic AI Hardware, Paper No. P-19
 October 2025 · Kitakyushu, Japan

Non-refereed Poster

[6] **Image Classification Using a Diffusion Model as a PreTraining Model**
 Kosuke Ukita, Xiaolong Ye, Tsuyoshi Okita
 arXiv preprint arXiv:2505.06890, pp. 1-10
 May 2025

Non-refereed Preprint

[7] **AURA-MFM: 詳細な行動理解のためのCLIPを用いたマルチモーダル基盤モデルの構築**
 松石康希, 浮田嵩祐, 大北剛
 情報処理学会研究報告(IPSJ SIG Technical Report), Vol. 2025-UBI-85, No. 8
 February 2025 · Osaka, Japan

Non-refereed Oral

[8] **Improving Human Activity Recognition through Diffusion Models**
 Kosuke Ukita, Tsuyoshi Okita
 The International Symposium on Applied Engineering and Sciences (SAES 2024), No. p0052
 November 2024 · Kitakyushu, Japan

Non-refereed Oral

[9] **Ensemble Learning of Models for Human Activity Recognition**
 Asahi Miyazaki, Kosuke Ukita, Tsuyoshi Okita
 The International Symposium on Applied Engineering and Sciences (SAES 2024), No. p0053
 November 2024 · Kitakyushu, Japan

Non-refereed Oral

[10]

Analysis of Human Activity Recognition by Diffusion Models

Kosuke Ukita

Tsuyoshi Okita

Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24), pp. 458–463

October 2024 · Melbourne, Australia

Refereed

Oral

[11]

表現条件付き潜在拡散モデルと表現学習

浮田嵩祐

Ye Xiaolong, 大北剛

情報処理学会研究報告, 火の国情報シンポジウム 2024, No. A4-4, pp. 1-8

March 2024 · Kagoshima, Japan

Non-refereed

Oral

[12]

Towards LLMs for Sensor Data: Multi-Task Self-Supervised Learning

Tsuyoshi Okita,

Kosuke Ukita

, Koki Matsuishi, Masaharu Kagiya, Kodai Hirata, Asahi Miyazaki

Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct), pp. 499–504

October 2023 · Cancun, Mexico

Refereed

Oral

GRANTS

- SPRING Scholarship Awardee

Japan Science and Technology Agency (JST)

April 2026 - March 2029 · JPY ¥1,800,000 (+ ¥900,000) / year
- Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

June 2026 · USD \$500
- International Travel Grant

The Telecommunications Advancement Foundation

April 2026 · JPY ¥290,000

AWARDS

- Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

June 2026
- Spotlight Presentation (Top 3%)

The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)

May 2026
- Promising Researcher Award

Information Processing Society of Japan (IPSJ) Kyushu Branch

March 2024
- Best Research Award

The Japan Statistical Society (JSS), Special Committee of Statistical Education (CSE), The 8th Sports Data Analysis Competition

May 2019

SKILLS

LANGUAGES

Python · C++ · C · Java · JavaScript · TypeScript · Go · SQL · R

FRAMEWORKS & TOOLS

PyTorch · TensorFlow · Hugging Face · Jupyter · ABCI · Docker · Git

OTHER SKILLS

Deep Learning · Neural Networks · Model Training · Data Processing · API Development · Linux

